

House Price Prediction

Project Summary:

Business Problem

Accurate property valuation is critical for real estate agencies, property investors, and home buyers. Traditional pricing methods often rely on manual assessment and market experience, which can lead to inconsistent valuations. This project aimed to develop a machine learning solution capable of predicting house prices based on property characteristics and identifying the key factors that drive property value.

Dataset and Approach

The project utilized a housing dataset containing property attributes such as area, bedrooms, bathrooms, stories, parking spaces, furnishing status, and other amenities. Data preprocessing included exploratory analysis, handling categorical variables through one-hot encoding, and preparing the dataset for model training.

To evaluate predictive performance, the dataset was split into training and testing sets (80:20). Two machine learning algorithms were implemented and compared:

- Linear Regression
- Random Forest Regressor

Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Key Insights

The analysis identified **area, number of bathrooms, parking availability, and preferred location** as the most influential factors affecting house prices. Properties with larger living spaces and better amenities consistently demonstrated higher market values.

Among the evaluated models, **Linear Regression achieved the best performance**, with an MAE of **970,043**, RMSE of **1,324,507**, and an **R^2 Score of 0.653**. In comparison, the Random Forest Regressor achieved an R^2 Score of **0.612**. These results indicate that Linear Regression provided more accurate and reliable predictions for this dataset.

Business Impact

The developed solution enables data-driven property valuation by providing consistent and scalable price estimates. The insights generated from the analysis can support pricing strategies, investment decisions, and market analysis within the real estate sector.

Recommendation

Real estate organizations should prioritize properties with larger areas, better amenities, sufficient parking, and desirable locations, as these characteristics have the strongest influence on market value. Integrating predictive analytics into the valuation process can improve pricing accuracy and support more informed business decisions.

Conclusion

This project demonstrated the practical application of machine learning in real estate analytics. By leveraging historical housing data, the model successfully predicted property prices and identified the key factors driving market value. The findings highlight the importance of data-driven decision-making in improving property valuation and supporting strategic business operations.